somehow both dead and alive. Yet we only ever see living cats or dead cats, not a superposition of them (whatever that might mean). Von Neumann wanted to avoid this problem, which is why he had been so explicit about the existence of wave function collapse in his book. But this still left the problem of how and why that collapse occurred.

Von Neumann's solution was to make the observer—whoever was looking—responsible for wave function collapse. "We must always divide the world into two parts, the one being the observed system, the other the observer," Von Neumann said. "Quantum mechanics describes the events which occur in the observed portion of the world, so long as they do not interact with the observing portion, with the aid of the [Schrödinger equation], but as soon as such an interaction occurs, i.e. a measurement, it requires the [collapse of the wave function]."

It's not entirely clear what von Neumann meant by this. Some took him to be saying that consciousness itself causes the collapse of the wave function; this was a view promoted by physicists Fritz London and Edmond Bauer in a book they wrote several years later, heavily influenced by von Neumann's work. Wigner also adopted this view later. But this is a weird view. Stating that consciousness collapses wave functions does arguably solve the measurement problem but only at the price of introducing new problems. How could consciousness cause wave function collapse? Since wave function collapse violates the Schrödinger equation, does that mean that consciousness has the ability to temporarily suspend or alter the laws of nature? How could this be true? And what is consciousness anyhow? Who has it? Can a chimp collapse a wave function? How about a dog? A flea? "Solving" the measurement problem by opening the Pandora's box of paradoxes associated with consciousness is a desperate move, albeit one that seemed reasonable at the time, in the absence of other fully developed solutions to the measurement problem.

Strange as it was, von Neumann may have also held the view that consciousness is responsible for wave function collapse. But he sidestepped this question in his textbook; there, he claimed that conscious observers held no special status in the theory. "The boundary between the [observer and observed] is arbitrary to a very large extent," he wrote. Sounding a positivist note, he claimed that "experience only makes statements of this type: an observer has made a certain (subjective) observation; and never